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Dear Professor Jed Kaplan,
Editor, Global and Planetary Change,

We are pleased to submit our manuscript entitled "*Hydroclimatic variability drove human-megafauna-environment interactions during the late Pleistocene/Early Holocene in Central Chile*" for your consideration at *Global and Planetary Change*.

Our study provides a novel perspective by explicitly linking paleohydrological transport mechanisms (fluvial, alluvial, and storm events) with ecosystemic responses and human behavior during the late Pleistocene and the onset of the Holocene in central Chile. Based on multi-proxy analyses from the archaeological and paleontological site of Taguatagua 3 in the central Chile valley, we demonstrate how atmospheric and hydrological variability—including ENSO-like conditions and shifts in the Southern Westerly Winds—drove early human settlement patterns and ecosystem dynamics in subtropical South America. Our research highlights the emergence of hydrological refugia as crucial zones of stability during abrupt climatic changes, ultimately aiding early human adaptation and survival.

A notable feature of our approach is the use of paleoclimate models (PMIP4/CMIP6) in conjunction with sedimentology, stable isotopes, and paleoecological evidence, creating a robust framework to correlate local sedimentary records with large-scale ocean-atmosphere dynamics. Our findings capture the role of jet stream shifts and major global events—including Heinrich Stadial 1, the Younger Dryas, and the Antarctic Cold Reversal—in shaping regional human–megafauna–environment interactions. This integrative perspective on climate teleconnections has broad implications for our understanding of past and present Coupled Human and Natural Systems.

We believe this manuscript will appeal to the diverse readership of *Global and Planetary Change*, as it provides an holistic lens for interpreting past climate-driven changes in human–environment interactions, a topic of urgent relevance in the face of ongoing climate challenges.

Thank you for your time and consideration.

Sincerely,

Matías Frugone-Álvarez

On behalf of all co-authors

Information to Ensure a Fair Review Process:

Originality: We confirm that this manuscript is original, unpublished, and not under consideration elsewhere.

Conflict of Interest: The authors declare no conflicts of interest that could have influenced the outcomes or interpretation of this study.

Data Availability: All data and code supporting this research will be made publicly available upon acceptance for reuse via Zenodo and Github.

Supplementary Materials: We have uploaded all additional data files and reference material to the auxiliary files section to facilitate the review process.

Thank you for considering our manuscript. We look forward to contributing to your journal and to engaging with the reviewer feedback.